

CompSci 234/NetSys 210
Advanced Topics in Networking
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Unit 1: The Internet and Layering Structure

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Slide adopted from Kurose/Ross and Prof. Venkatasubramanian

Agenda

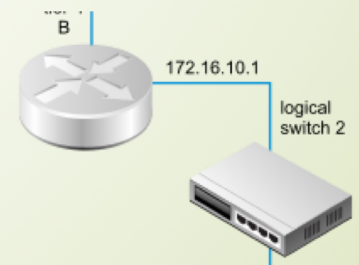
- What is the Internet
- Network Edge
- Network Core: Circuit Switched vs. Packet Switched
- Protocol Layers and UNIX Sockets
- History and Internet design philosophy

Network Evolution

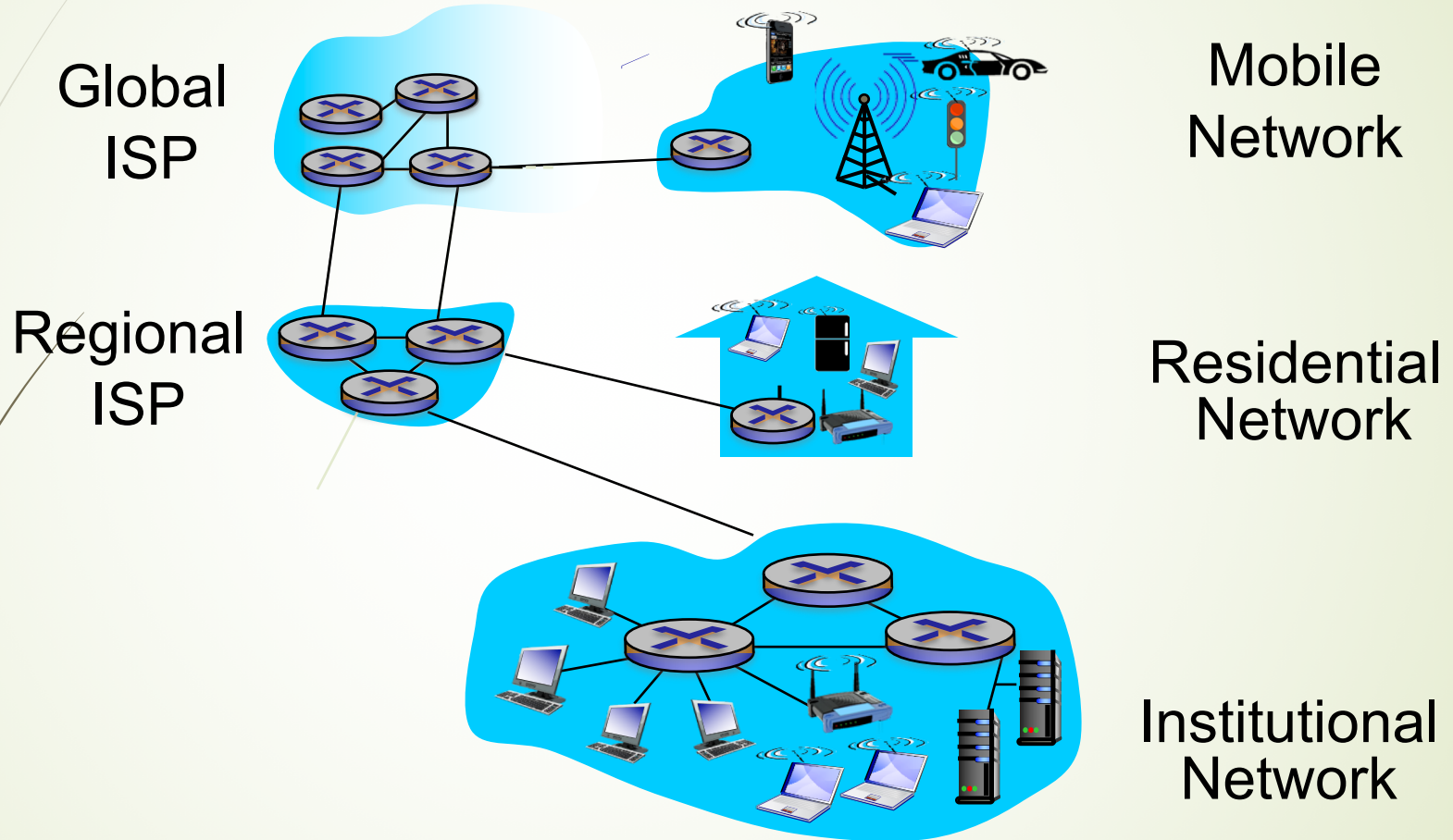
- 1st Gen (Circuit Switching), **Telephone** : **wires**
 - Running a pair of wires to every home & office
 - Dynamically constructing a path from caller to callee
- 2nd Gen (Packet Switching), **The Internet:** **computing devices**
 - Data sent in independent chunks
 - Each chunk contains the name of the final destination.
- 3rd Gen, **Information-Centric Network?** : **data**
 - Don't care where the data are from
 - Supported by the network infrastructure? Or by applications?

Internet Consists of

- Too many Connected computing devices:
 - hosts = end systems
 - running network apps
- Communication links
 - fiber, copper, radio, satellite
 - transmission rate: bandwidth
- Packet switches: forward packets (chunks of data)
 - routers and switches



Internet: Network of Networks



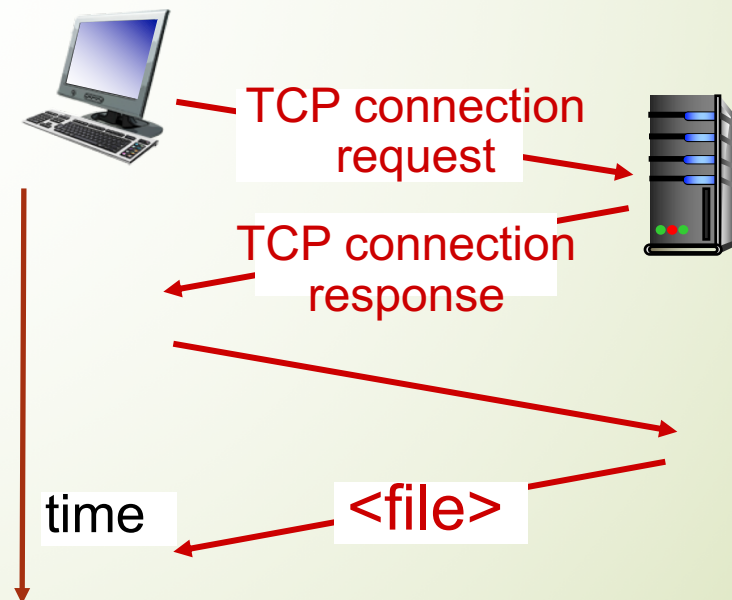
➡ Interconnected ISPs

Software Components in the Internet

- Internet standards
 - RFC: Request for comments
 - IETF: Internet Engineering Task Force
- protocols control sending, receiving of messages
 - e.g., HTTP, TCP, IP, 802.11

What is a Protocol

- ▶ protocols define **format**, **order** of messages **sent** and **received** among network entities, and **actions taken** on message transmission, receipt
- ▶ all communication activity in Internet governed by protocols



The Internet Provides Services

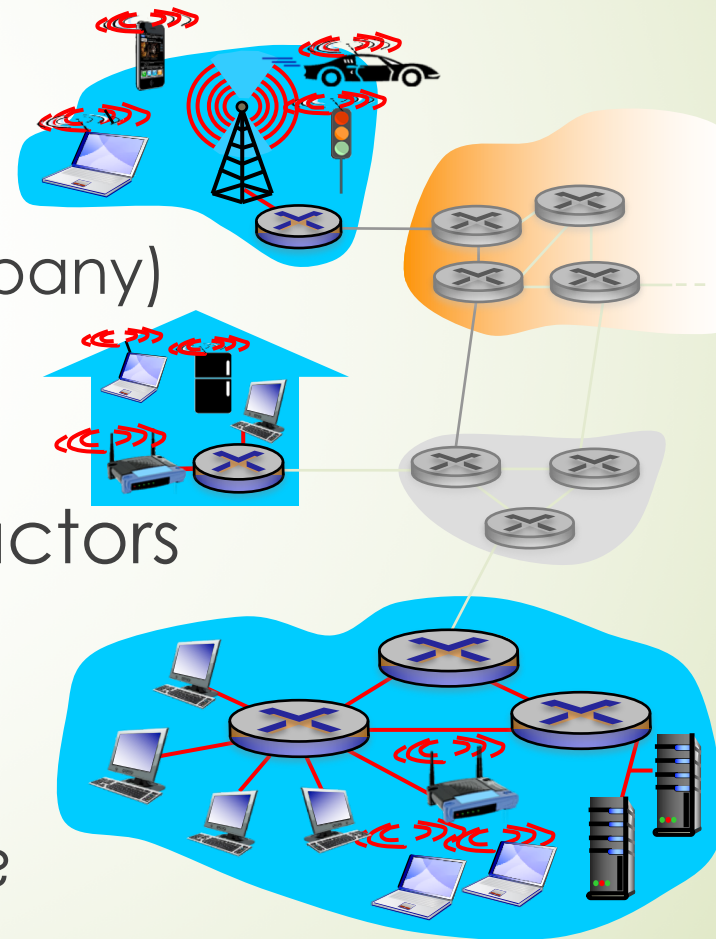
- **Infrastructure** that provides services to applications:
 - Web, VoIP, email, games, e-commerce, social networks, ...
- **Programming interface** to apps
 - hooks that allow sending and receiving app programs to “connect” to Internet: **socket APIs**
 - provides service options, analogous to postal service: **different classes**

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Access Networks

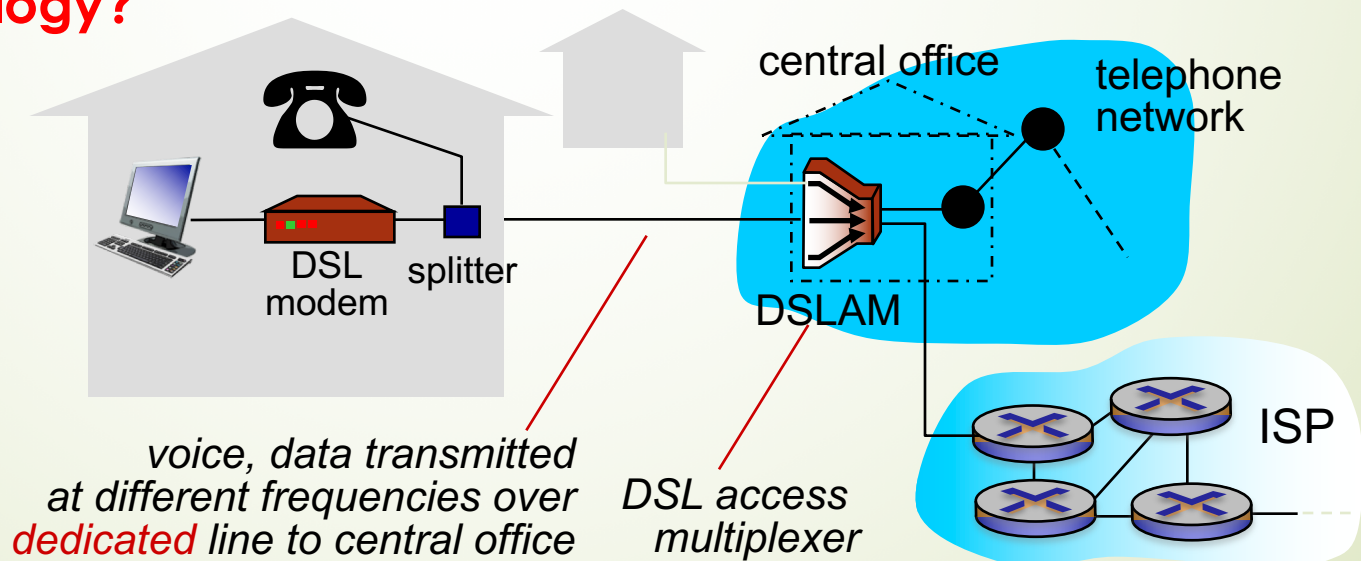
- Q: How to connect end systems to edge router?
 - **residential** access nets
 - **institutional** access networks (university, company)
 - **mobile** access networks (T-Mobile, A&T, Verizon..)
- Q: Main performance factors
 - **bandwidth** (bits per second) of access network?
 - **shared** versus **dedicated**?



Sample Wired Access Network: ADSL

- use existing telephone line to central office DSLAM
- data over DSL phone line goes to Internet
- voice over DSL phone line goes to telephone net

Q: Can you come up with a shared wired access network technology?



Two Types of Wireless Access Networks

- shared air medium with wireless hosts
- via base stations, or access points

wireless LANs:

- within building (100 ft.)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate



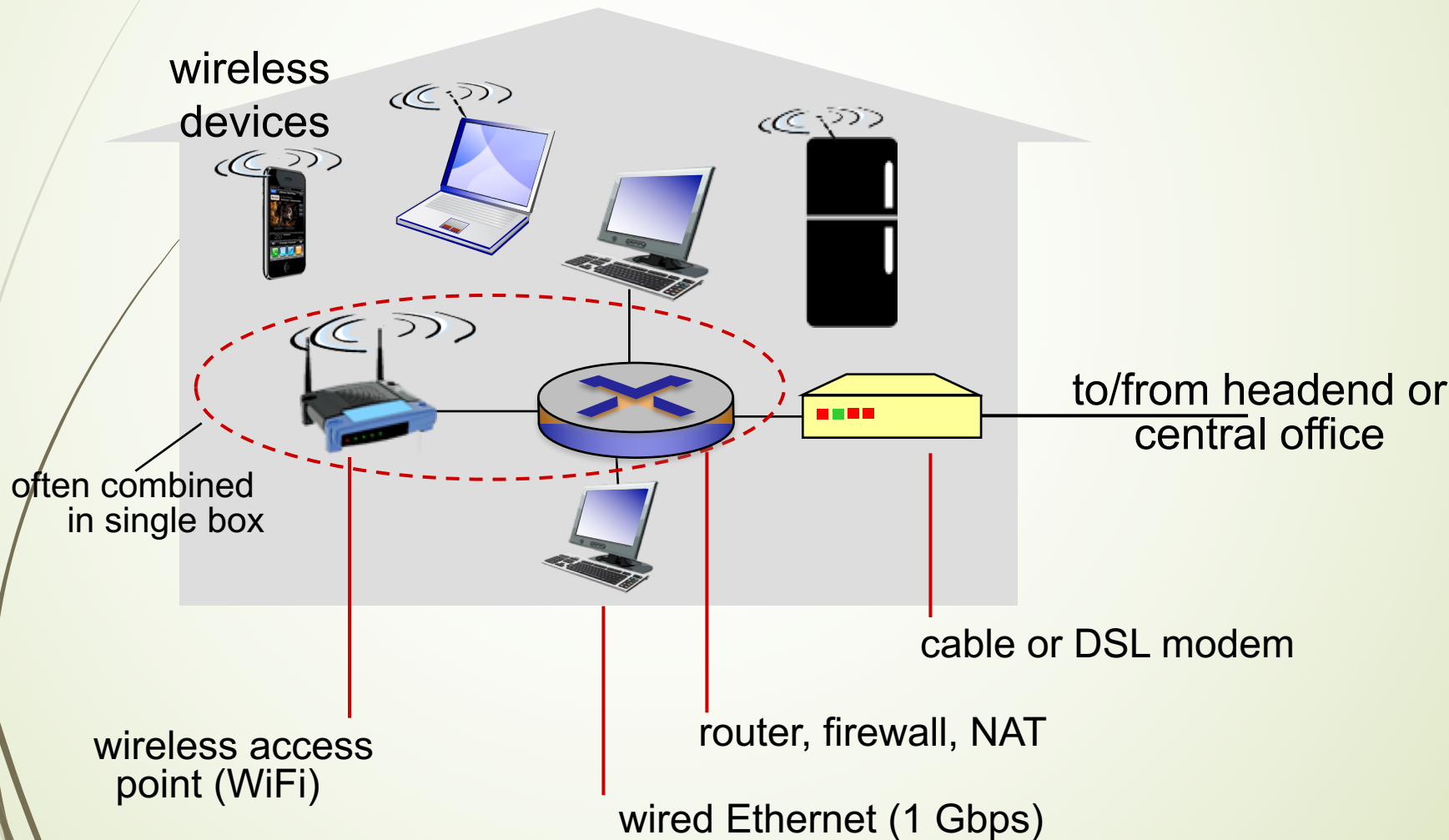
to Internet

wide-area wireless access

- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE



Home Networks



Enterprise Access Networks

- Typically used in companies, universities, etc.
 - 10 Mbps, 100Mbps, 1Gbps, 10Gbps Ethernet

